

CUSTOMER CHURN ANALYSIS REPORT

1 Executive Summary

This project focuses on analyzing and predicting customer churn for a telecommunications company. The primary objective is to identify key drivers of customer attrition and build a predictive model to assist the Customer Success team in taking proactive retention measures.

- **Overall Churn Rate:** ~27%
- **Predictive Model Accuracy:** 84.95% (Random Forest Classifier).
- **High-Risk Profile:** Customers on a Month-to-month contract, utilizing Fiber Optic internet, and having a high volume of technical support tickets.

2 Methodology

The project was executed through a rigorous three-stage pipeline:

1. **SQL (Data Exploration):** Utilized SQL queries to identify lost VIP customers, aggregate churn rates by contract type, and analyze technical support behavior patterns.
2. **Python (Machine Learning):** Performed data preprocessing, handled class imbalance, and trained a *Random Forest* model to predict churn probability. Successfully isolated 56 current customers with a high risk of churning.
3. **Power BI (Visualization):** Designed an interactive dashboard to track critical KPIs and provide a clear view of the high-risk customer segments.

3 Key Insights

Based on the data exploration and visualization, three primary insights were discovered:

3.1 Contract Type Vulnerability

Customers on **Month-to-month** contracts exhibit a significantly higher churn rate compared to those on 1-year or 2-year commitments. This indicates a lack of long-term loyalty and high sensitivity to service or pricing changes within this group.

3.2 Technical Issues (Fiber Optic & Tech Tickets)

The data reveals that customers using **Fiber Optic** services have an unusually high churn rate. Furthermore, SQL analysis highlighted that customers who opened **3 or more Technical Support Tickets** possess a churn probability of over 50%. This strongly suggests underlying infrastructure instability or an inefficient incident resolution process.

3.3 Customer Tenure

The highest risk of churn occurs during the **first 10 months** of the customer lifecycle. Once a customer surpasses the 24-month mark, brand loyalty increases significantly, and the churn rate drops.

4 Business Recommendations

To address the identified issues and mitigate revenue loss, I propose the following strategic actions:

- **Proactive Retention Campaign:** Deploy the list of 56 high-risk customers (identified via the Machine Learning model with $> 60\%$ churn probability) to the Telesales team immediately. Offer personalized discounts or service upgrades to retain them before their next billing cycle.
- **Contract Upgrade Incentives:** Launch a targeted marketing campaign encouraging Month-to-month customers to switch to an annual contract. For instance, offer one month of free service or a complimentary streaming add-on upon signing a 1-year agreement.
- **Operational Improvements:** Conduct a thorough technical audit of the Fiber Optic infrastructure in high-churn areas. Additionally, implement a "Red Flag" system: if a customer opens a 3rd tech ticket within a single month, escalate the issue directly to a Tier-2 technical lead to ensure rapid resolution.

5 Conclusion

By integrating traditional data exploration (SQL), predictive analytics (Machine Learning in Python), and modern data visualization (Power BI), this project delivers a comprehensive solution to customer churn. It not only explains *why* customers leave but also predicts *who* is most likely to leave next, providing a clear roadmap for operational optimization and revenue protection.